

Developing an XR-Integrated Digital Twin for Hydrogen Pipeline Monitoring and Navigation

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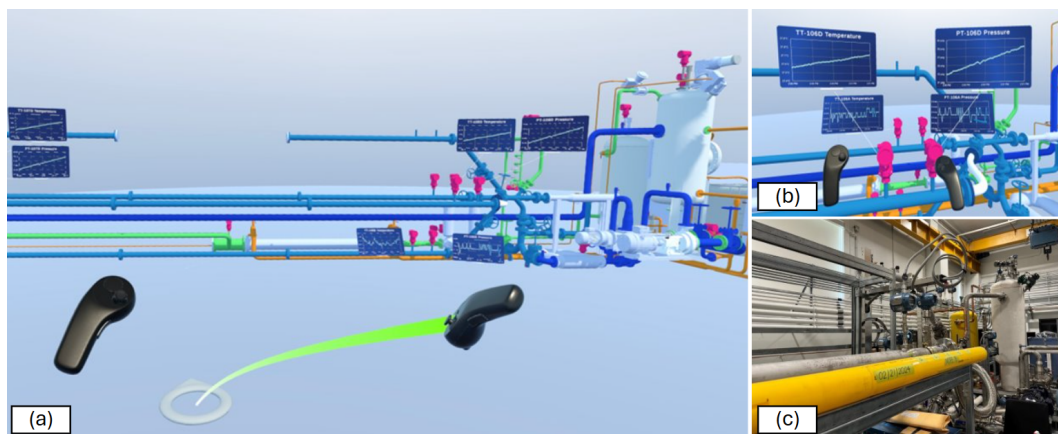


Figure 1: XR-based Digital Twin (DT) of a hydrogen pipeline facility: (a) an immersive XR environment featuring teleportation, pipeline models, and real-time data visualization graphs; (b) a graph-based visualization of real-time sensor data from the physical facility; (c) the physical pipeline facility with sensors and actuators for hydrogen pipeline research and innovation.

ABSTRACT

This paper presents an immersive extended reality (XR) based digital twin (DT) system designed for a hydrogen pipeline test facility, combining real-time data visualization with interactive virtual navigation. Employing 3D pipeline models and a point-and-teleport locomotion method with orientation specifications, the system enables efficient exploration of the physical pipeline facility's virtual counterpart. Preliminary feedback from field experts highlights the system's potential to provide effective navigation and reduce cognitive fatigue through the visualization of the sensor data.

Index Terms: Extended Reality, Digital Twin, Pipeline, Data Visualization, Immersive Teleportation, Monitoring

1 INTRODUCTION

In the oil and gas pipeline industry, many studies have highlighted the benefits of digital twin (DT) technology in addressing critical challenges, such as the critical role of DTs in supporting the life cycles of large-scale facilities and enhancing decision-making, operational efficiency, and safety [4, 5]. However, there is a notable lack of research exploring the use of extended reality (XR) integrated DTs for monitoring, simulating, and operating pipelines [3]. This is particularly evident in the context of user interaction and human-centered studies within XR environments, which are critical for leveraging the full capabilities of these technologies.

Understanding how users navigate simulated pipeline environments and interact with information within an immersive digital twin is crucial. This knowledge helps assess pipeline conditions and supports timely, informed decision-making. These navigation capabilities are particularly crucial for the safe and efficient management of pipelines, especially in scenarios involving complex, real-time data integration and monitoring. Various locomotion methods have been proposed to improve the users' navigation performance and experience in XR. Bozgeyikli et al. proposed a technique that integrates point and teleport (P&T) with orientation specification in one step [1]. Mori et al. then compared the P&T technique, where a line of trajectory cast from a handheld device to suggest where to teleport in the immersive environment, with the P&T plus orientation specification technique and found they have similar performance in task completion time and accuracy [2].

In this paper, we describe the development of an XR-based pipeline DT simulation tailored for hydrogen pipeline facilities, which adopts the P&T locomotion method with orientation specification to enhance user navigation and situational awareness in the immersive environment. We present preliminary feedback from potential stakeholders, highlighting the system's capabilities, challenges, and future directions for practical implementation.

2 IMMERSIVE XR-BASED PIPELINE DIGITAL TWIN

Our project focuses on the development of an immersive XR-based DT environment (Figure 1(a)) for a hydrogen pipeline test facility. The DT connects to the Advanced Pipeline Research and Innovation Laboratory (APRIL), a physical pipeline facility equipped with over 70 industrial sensors and actuators that collect pressure, temperature, and flow rate information across the system (Figure 1(c)).

The immersive XR environment, developed in Unity (2022.3 LTS) and deployed on a Meta Quest 3 head-mounted display (HMD), was designed based on input from pipeline engineering experts from both academia and industry. This environment functions as a remote control center, reducing physical contact during

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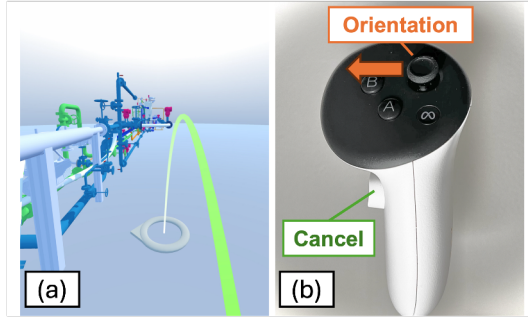


Figure 2: Point and teleportation (P&T) with orientation specifications: (a) a teleportation ray with a directional reticle, (b) the thumbstick-based orientations and the grip button for cancellation.

pipeline operations. To achieve this, detailed CAD models of the APRIL facility were created and imported into Unity as Filmbox (FBX) files after optimization to preserve the object hierarchy and reduce polycount, improving performance.

For real-time data streaming between the APRIL facility and the DT model, the system employs the Open Platform Communication Unified Architecture (OPC UA) protocol, facilitated by a commercially available OPCUA4Unity Unity asset. A custom script subscribes to the unique node IDs for individual sensors to retrieve live data that was displayed as graphs within the XR environment using the Graph and Chart Unity asset (Figure 1(b)). These graphs represent the real-time pressure, flow rate, and temperature data for monitoring purposes.

This XR-integrated DT for the APRIL facility serves as a proof of concept with the potential for adoption in large-scale immersive DT projects for monitoring, control, situational awareness, and decision-making.

3 NAVIGATING XR PIPELINE DIGITAL TWIN ENVIRONMENT

The pipeline facility occupies a large physical footprint with numerous points of interest, requiring users to have the means to navigate its DT to visualize and interact with all components in the XR environment. Especially when pipeline engineers detect a potential leakage, they need to navigate through and inspect a series of sensors within a short period of time (Figure 1(b)). Thus, being able to smoothly navigate the facility in the XR environment and look at the direction of the sensor data visualization is essential to enhance the efficiency of understanding the pipeline states and determining the leakage status. To facilitate navigation, we implemented P&T method as the primary locomotion method, rather than smooth locomotion (free movement). In teleportation mode, users aim the controller to select a destination, with a curved parabolic ray extending outward as a visual cue. At the end of the ray, a circular marker displays an arrow indicating the user's selected post-teleport orientation, which is adjusted by moving the thumbstick in the desired direction (Figure 2). To abort the teleport action, users can press the controller's grip button. Otherwise, releasing the thumbstick back to its neutral position confirms the teleport. Users can toggle teleportation mode on or off by clicking the controller thumbstick, reducing visual clutter in the interface when the feature is not needed.

4 PRELIMINARY FEEDBACK AND FUTURE WORK

Following a user-centered design process, we regularly meet with field experts such as pipeline engineers and researchers specialized in leakage detection. Each meeting session has a duration of around one hour and involve three to five field experts. During the session, notes were taken by investigators to gather their feedback on the system and learn what functionality and in which context they want to use the XR-based DT system. Other qualitative feedback, such

as current challenges in the pipeline monitoring process, are also collected. Field experts generally provided positive feedback on our XR-based DT system that enables real-time sensor data to be visualized through both graphs and spatial representation in XR, and acknowledged this could improve situational awareness and decision-making, ultimately enhancing safety and operational efficiency in hydrogen pipeline monitoring. One thing pointed out by a field expert in the industry is that in the real world, monitoring the pipeline facility is an ongoing task that requires the pipeline engineer to examine real-time data from a massive number of sensors continuously, which causes significant cognitive fatigue. To address this, we plan to integrate an enhanced real-time transient model and other types of AI-based leak detection and prediction methods into our system so that users in a virtual environment only need to look at sensor data visualization when there is a risk of leakage, potentially reducing the cognitive load of the users.

We observed that field experts were all able to teleport easily in the XR-DT environment after being given a short introduction to how to use the P&T technique with orientation specifications. They perceived that our locomotion technique is efficient for navigating the DT and monitoring the sensor data while sitting in the office. They also provided positive feedback on being able to look at the part of the facility they wanted to inspect once they teleported using the simultaneous orientation-specific technique. However, they mentioned that the visual cue of the orientation specification could be improved to easily recognize which direction the arrow was pointing if they wanted to teleport to target positions that were far away or obstructed by the facility. From preliminary feedback, we found that integrating the P&T technique with orientation specification has the potential to enhance navigation in complex virtual spaces with obstacles and many target objects to look at. However, proper visual cues need to be selected for orientation specification so that the technique can be applied to practical scenarios.

5 CONCLUSION

We proposed an XR-integrated DT system for a hydrogen pipeline for monitoring and navigating the facility. Overall, field experts provided positive feedback with important requirements on the features that need to be implemented for practical use. The XR-based DT of the APRIL is designed to simulate real-world scenarios at a manageable scale to test and refine the technology. In the future, this approach can be extended to real-world systems, to develop large-scale DTs for monitoring, operational control, and immersive training applications to support field technicians.

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